Ultra-LSNT: Edge-Native Sparse Expert Framework for Wind Power Forecasting under Corrupted SCADA

An integrated chain from data degradation, sparse forecasting, and feasibility guidance to dispatch screening

Data Layer

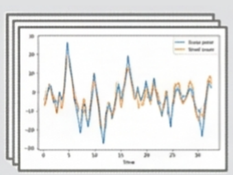
Wind Farm SCADA Historical Window

Multivariate Inputs:
Wind Speed, Power, Environmental Variables

Corruption Types

Data Degradation Injection

- Gaussian Noise
- Drift Bias
- Quantization Error
- Flatline Fault
- Spike Outlier
- Correlated Missingness





Industrial field

Forecast Layer

Moving Average Decomposition

Trend Branch → Lightweight Linear head

Seasonal / Residual Branch

Dual-Convolution Multi-Scale Embedding

Short-Term Convolution $\mathbf{k_s = 3}$ → Mid-Term Convolution $\mathbf{k_m = 7}$ → Fusion Projection **E**

Sparse MoE Encoder

Router softmax → Top-k Selection

Experts: E_1 E_2 E_3 · · · EN

Jump Gate Conditional Skip

Weighted Aggregation

24-Step Multi-Horizon Wind Power Forecast Trajectory

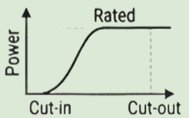
Feasibility Guidance and Interface Layer

FG-MoE Feasibility Guidance

Deployable Wind Speed Proxy

Persistence + Short-Term Mean + Local Trend

Wind Turbine Power Envelope



Physical Constraint Loss
Training discourages physically infeasible

Fixed Shared Interface

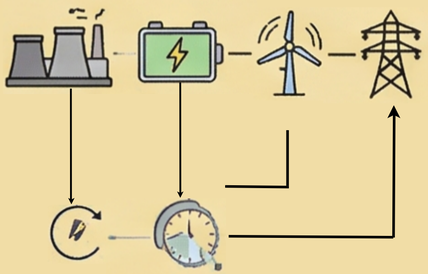
Affine Mapping +
+
Deterministic Clipping

Shared by all models than tuned per model

Dispatch-Usable Forecast Trajectory

Dispatch / Value Layer

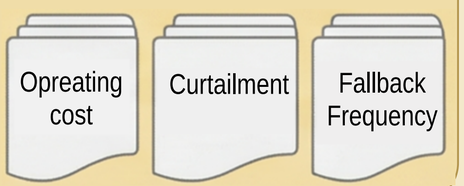
Day-Ahead Dispatch Module



24-Hour Rolling **Evaluation**

1-Hour Aggregation

Value Assessment



RTS-24 / DC-OPF Coarse Screening

Simplified network-constrained validation step

Robustnessunder corrupted inputs

Low latency and low memory for edge deployment

Dispatch usability under a shared interface